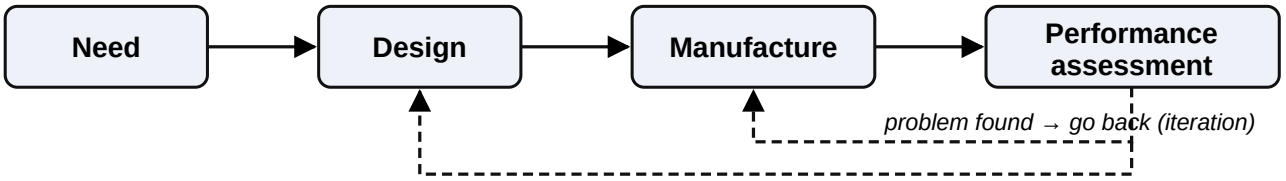
Question 1 — 20 points

INTRODUCTION TO THE DESIGN PROCESS

(a) Sequential stages of the mechanical design process (6 pts)



1. **Identification of a need:** design always starts from a need, e.g. a problem to solve or a product to improve. It is turned into a clear problem statement with the required functions, loads, constraints, cost and so on.
2. **Design:** the engineer generates concepts, chooses materials and geometry, and does the **analysis** (stress, deflection, failure theories, factor of safety). The design is then evaluated and optimized, and the result is drawings and specifications.
3. **Manufacture:** the part is produced from the design, using the chosen processes, tolerances and assembly.
4. **Performance assessment:** the manufactured part is tested and checked in service to see whether it meets the original need (strength, life, function, safety).

(b) Why the design process is iterative (6 pts)

The process is **iterative** because the first design is rarely the best one. Information that is only available later, such as manufacturing difficulties, test results or service behaviour, often shows weaknesses that could not be fully predicted at the start. Each stage feeds back into the earlier ones, and the cycle repeats until the product meets the need.

If a problem is identified:

- **During manufacturing:** go back to the **design** stage and modify the design (geometry, tolerances, material, process) so that it can be manufactured.
- **During performance assessment:** find the cause, then go back to **manufacture** (if the fault is in production quality) or to **design** (if the design itself is inadequate). Re-analyse, correct it and assess again.

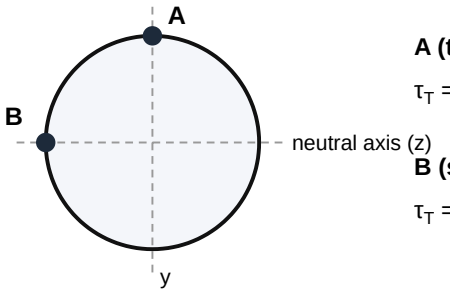
(c) Five mechanical failure modes (8 pts)

Failure mode	Physical meaning
1. Yielding	The stress goes above the yield strength S_y, so the part deforms **permanently (plastically)** and no longer keeps its shape or function. This is the main failure mode for ductile materials under static load.
2. Fracture	The part **breaks into pieces** when the stress reaches the ultimate strength. It is typical of brittle materials, which fail suddenly with little deformation.
3. Buckling	A slender member under **compression** suddenly **bends sideways** (loses stability) at a critical load, even though the stress may still be below S_y.
4. Creep	Slow, **time-dependent permanent deformation** under a constant load, mainly at **high temperature**. It can end in rupture.
5. Fatigue	Failure under **repeated or fluctuating (dynamic) loads**. A crack starts and grows over many cycles at stresses **below** the static strength.
(extra) Chemical failure – corrosion / wear	Material is lost from the surface by chemical attack (corrosion) or by rubbing contact (wear). This reduces the section and the strength over time.

Question 2 **20 points**

LOADED CANTILEVER CYLINDER

Given: L = 250 mm, d = 50 mm → c = d/2 = 0.025 m, A = 1.96×10^{-3} m², I = 3.06×10^{-7} m⁴, J = 6.12×10^{-7} m⁴
Axial **compressive** load F = 1 kN, transverse load V = 3 kN, torque T = 1 kN·m
Critical section: the fixed end (wall), where the bending moment is largest. Point **A** is the top fiber; point **B** is the side, on the neutral axis.



A (top): $\sigma_{bending}$ = max, $\tau_V = 0$
τ_T = max (outer surface)

B (side): $\sigma_{bending} = 0$, τ_V = max
τ_T = max (outer surface)

(a) Axial normal stress and maximum bending stress

Axial stress (compressive, uniform over the section, same at A and B):

$$\sigma_{axial} = -F/A = -(1000\text{ N})/(1.96\times10^{-3}\text{ m}^2) = -5.10\times10^{5}\text{ Pa}$$

$\sigma_{axial} = -0.51$ MPa (compression)

Maximum bending moment at the fixed end:

$$M_{max} = V\cdot L = (3000\text{ N})(0.25\text{ m}) = 750\text{ N}\cdot\text{m}$$

Maximum bending stress at the top fiber (point A). The load acts downward, so the top fiber is in tension:

$$\sigma_b = M\cdot c / I = (750)(0.025) / (3.06\times10^{-7}) = 6.13\times10^{7}\text{ Pa}$$

$\sigma_{b,max} = 61.27$ MPa (tension at A); $\sigma_b = 0$ at B (neutral axis)

(b) Transverse and torsional shear stresses at A and B

Transverse shear (it is zero at the top surface and largest at the neutral axis):

$\tau_{V,A} = 0$ (free top surface)

$$\tau_{V,B} = 4V / 3A = 4(3000) / [3(1.96\times10^{-3})] = 2.04\times10^{6}\text{ Pa}$$

$\tau_{V,A} = 0$ $\tau_{V,B} = 2.04$ MPa

Torsional shear (largest on the outer surface, so the same at A and B):

$$\tau_T = T\cdot c / J = (1000\text{ N}\cdot\text{m})(0.025\text{ m}) / (6.12\times10^{-7}\text{ m}^4) = 4.085\times10^{7}\text{ Pa}$$

$\tau_{T,A} = \tau_{T,B} = 40.85$ MPa

(c) Combined normal and shear stresses at A and B

Point	Normal stress $\sigma_x = \sigma_{axial} + \sigma_b$	Shear stress $\tau = \tau_V + \tau_T$
A	−0.51 + 61.27 = **60.76 MPa**	0 + 40.85 = **40.85 MPa**
B	−0.51 + 0 = **−0.51 MPa**	2.04 + 40.85 = **42.89 MPa**

A: σ_x = 60.76 MPa, τ = 40.85 MPa | B: σ_x = −0.51 MPa, τ = 42.89 MPa

At B the transverse and torsional shear stresses act along the same line (the y direction), so they add, as in the course notes.

> **Note:** the lecture notes show $\tau_T \approx 0.04$ MPa (so 2.08 MPa at B). That comes from using T = 1 N·m. With T = 1 kN·m = 1000 N·m, as given, the correct value is **40.85 MPa**. With the exact A and I instead of the rounded given values, you get −0.509 MPa and 61.1 MPa; both versions are accepted.

Question 3 **20 points**

THREE-DIMENSIONAL STRESS STATE

Given: $\sigma_x = 20$ MPa, $\sigma_y = -10$ MPa, $\sigma_z = 15$ MPa, $\tau_{xy} = 5$ MPa; $\tau_{yz} = \tau_{xz} = 0$

(a) Full 3-D stress tensor

$$[\sigma] = \begin{vmatrix} \sigma_x & \tau_{xy} & \tau_{xz} \\ \tau_{xy} & \sigma_y & \tau_{yz} \\ \tau_{xz} & \tau_{yz} & \sigma_z \end{vmatrix} = \begin{vmatrix} 20 & 5 & 0 \\ 5 & -10 & 0 \\ 0 & 0 & 15 \end{vmatrix} \text{ MPa}$$

(b) Characteristic equation and principal stresses

$$\det([\sigma] - \sigma[I]) = \begin{vmatrix} 20-\sigma & 5 & 0 \\ 5 & -10-\sigma & 0 \\ 0 & 0 & 15-\sigma \end{vmatrix} = 0$$

Expanding along the third row:

$$(15-\sigma)\cdot[(20-\sigma)(-10-\sigma) - 5^2] = 0$$

$$(15-\sigma)\cdot(\sigma^2 - 10\sigma - 225) = 0$$

As a cubic ($\sigma^3 - I_1\sigma^2 + I_2\sigma - I_3 = 0$):

$$I_1 = 20 - 10 + 15 = 25 \quad I_2 = (20)(-10) + (-10)(15) + (15)(20) - 5^2 = -75 \quad I_3 = \det[\sigma] = -3375$$

$$\boldsymbol{\sigma^3 - 25\sigma^2 - 75\sigma + 3375 = 0}$$

Roots: $15 - \sigma = 0$ gives $\sigma = 15$. The quadratic gives $\sigma = [10 \pm \sqrt{(100 + 900)}]/2 = 5 \pm \sqrt{250} = 5 \pm 15.81$.

$\sigma_1 = 20.81$ MPa $\quad \sigma_2 = 15.00$ MPa $\quad \sigma_3 = -10.81$ MPa

Check: $\sigma_1 + \sigma_2 + \sigma_3 = 25 = I_1$ ✓ $\quad \sigma_1\sigma_2\sigma_3 = -3375 = I_3$ ✓

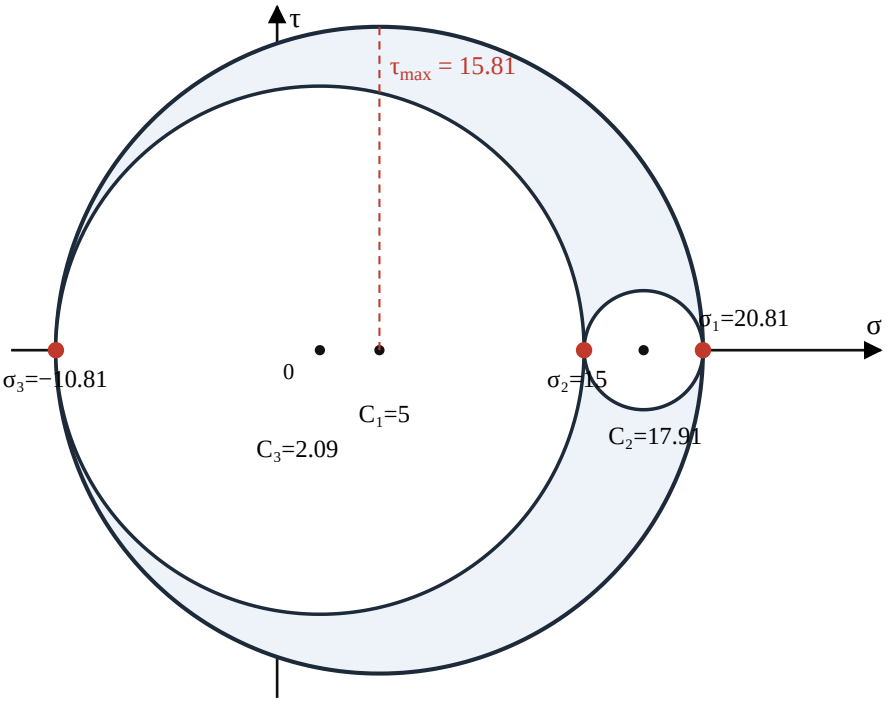
(c) Absolute maximum shear stress

$$\tau_{max} = (\sigma_1 - \sigma_3)/2 = (20.81 - (-10.81))/2$$

$\tau_{max} = 15.81$ MPa

(d) The three Mohr circles

Circle	Through	Center $C = (\sigma_i+\sigma_j)/2$	Radius $R = \lvert\sigma_i-\sigma_j\rvert/2$
1 (largest)	σ_1, σ_3	$(20.81 - 10.81)/2 =$ **5.00 MPa**	$(20.81 + 10.81)/2 =$ **15.81 MPa**
2	σ_1, σ_2	$(20.81 + 15)/2 =$ **17.91 MPa**	$(20.81 - 15)/2 =$ **2.91 MPa**
3	σ_2, σ_3	$(15 - 10.81)/2 =$ **2.09 MPa**	$(15 + 10.81)/2 =$ **12.91 MPa**



Drawn to scale (1 MPa = 10 units). The big circle (σ_1–σ_3) holds the two small ones, and its radius is τ_{max}.

Solved with **τ_{xy} = 5 MPa**, as written in the question. In the lecture example the 5 MPa was τ_{yz}. That version gives σ_1 = 20, σ_2 = 15.96, σ_3 = −10.96 MPa and τ_{max} = 15.48 MPa.

Question 4 **20 points**

FAILURE CRITERIA AND FACTOR OF SAFETY

Given: $\sigma_x = 40$ MPa, $\sigma_y = 30$ MPa, $\tau_{xy} = 20$ MPa, $\tau_{yz} = 10$ MPa; $\sigma_z = \tau_{xz} = 0$; $S_y = 100$ MPa (ductile)

(a) Principal stresses

$$[\sigma] = \begin{vmatrix} 40 & 20 & 0 \\ 20 & 30 & 10 \\ 0 & 10 & 0 \end{vmatrix} \text{ MPa} \Rightarrow \begin{vmatrix} 40-\sigma & 20 & 0 \\ 20 & 30-\sigma & 10 \\ 0 & 10 & -\sigma \end{vmatrix} = 0$$

$(40-\sigma)[(30-\sigma)(-\sigma) - 10^2] - 20[20(-\sigma) - 10\cdot 0] = 0$

$(40-\sigma)(\sigma^2 - 30\sigma - 100) + 400\sigma = 0$

$I_1 = 40 + 30 + 0 = 70 \quad I_2 = (40)(30) - 20^2 - 10^2 = 700 \quad I_3 = \det[\sigma] = -4000$

$\sigma^3 - 70\sigma^2 + 700\sigma + 4000 = 0$

Solving the cubic (calculator or trial and error):

$\sigma_1 = 56.31$ MPa $\quad \sigma_2 = 17.71$ MPa $\quad \sigma_3 = -4.01$ MPa

Check: sum = 70.0 = I_1 ✓ product = −4000 = I_3 ✓

(b) Maximum shear stress

$\tau_{max} = (\sigma_1 - \sigma_3)/2 = (56.31 + 4.01)/2$

$\tau_{max} = 30.16$ MPa

(c) Factor of safety – Tresca (maximum shear stress)

$|\sigma_1-\sigma_2| = 38.60, \quad |\sigma_2-\sigma_3| = 21.72, \quad |\sigma_1-\sigma_3| = 60.32$ MPa

$n = S_y / \max|\sigma_i-\sigma_j| = S_y / (\sigma_1-\sigma_3) = 100 / 60.32$

$n_{Tresca} = 1.66$

(d) Factor of safety – von Mises (distortion energy) and comparison

$\sigma' = \sqrt{\{ \tfrac{1}{2}[(\sigma_1-\sigma_2)^2 + (\sigma_2-\sigma_3)^2 + (\sigma_3-\sigma_1)^2] \}}$

$= \sqrt{\{ \tfrac{1}{2}[38.60^2 + 21.72^2 + 60.32^2] \}} = \sqrt{\{ \tfrac{1}{2}[1490.0 + 471.7 + 3638.4] \}} = \sqrt{2800} = 52.92$ MPa

$n = S_y / \sigma' = 100 / 52.92$

$\sigma' = 52.92$ MPa $\quad n_{von\ Mises} = 1.89$

Check from the components: $\sigma' = \sqrt{\{\tfrac{1}{2}[(\sigma_x-\sigma_y)^2 + \sigma_y^2 + \sigma_x^2 + 6(\tau_{xy}^2 + \tau_{yz}^2)]\}} = \sqrt{\{\tfrac{1}{2}[100 + 900 + 1600 + 3000]\}} = \sqrt{2800}$ ✓

Comparison:

- Both factors of safety are greater than 1, so the component is **safe (it does not yield)**.

- Tresca gives the **lower** factor of safety (1.66 < 1.89), about 12 % lower, so **Tresca is more conservative**. Its hexagon lies inside the von Mises ellipse (Tresca assumes $S_{sy} = 0.5S_y$, von Mises gives $S_{sy} = 0.577S_y$).
- Von Mises agrees better with test results for ductile materials, so it is more accurate. Tresca is the safer and simpler choice.

Question 5 **20 points**

SHAFT DESIGN UNDER COMBINED LOADING

Given: simply supported shaft, total length 200 + 600 + 200 = 1000 mm
F = 40 kN at x = 0.2 m and at x = 0.8 m (so a = 0.2 m, ℓ = 0.6 m); T = 40 kN·m at each load point, in opposite directions
S_y = 400 MPa, n = 2, von Mises criterion, stress concentrations neglected

(a) Support reactions and bending-moment diagram

$\Sigma M_A = 0$: $R_B(1.0) - 40(0.2) - 40(0.8) = 0 \Rightarrow R_B = 40$ kN

$\Sigma F_y = 0$: $R_A + R_B - 40 - 40 = 0 \Rightarrow R_A = 40$ kN

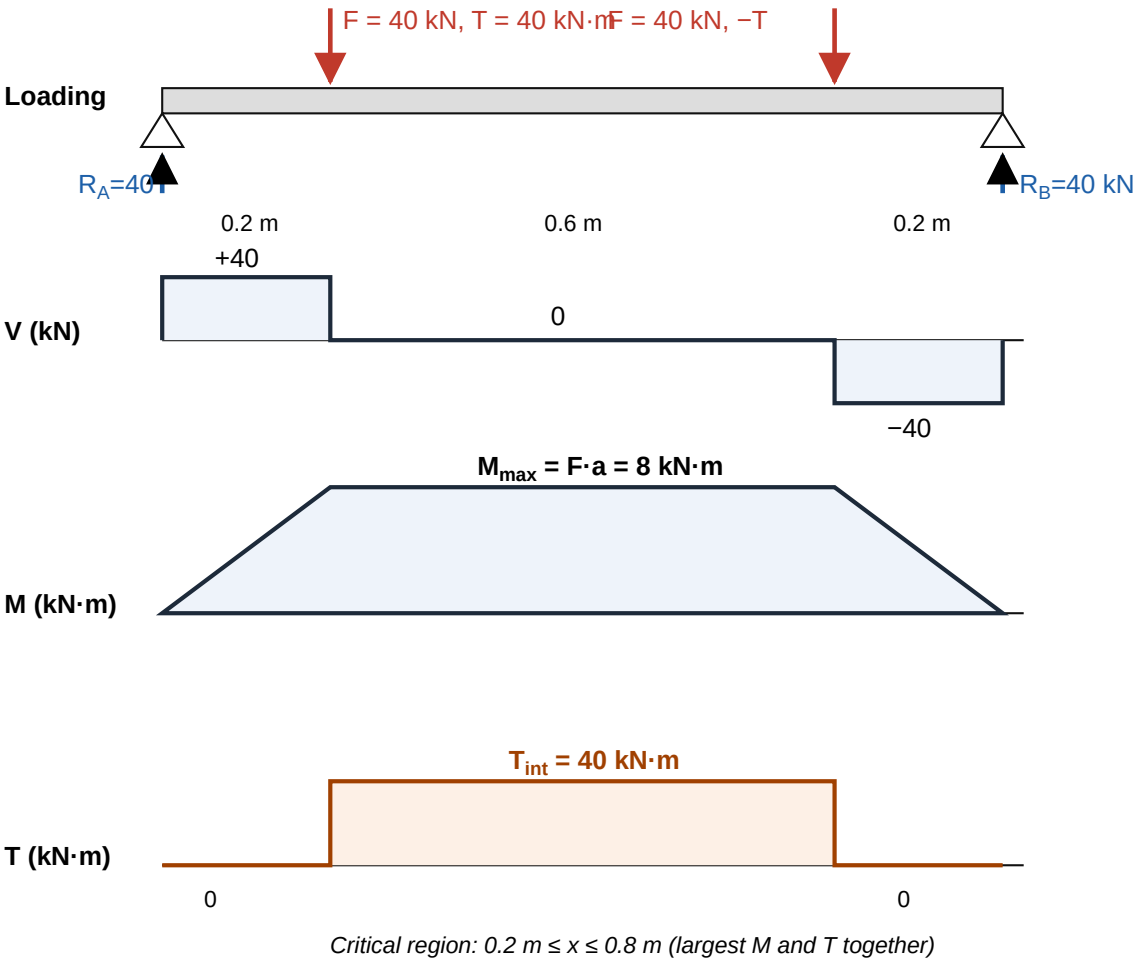
$R_A = R_B = 40$ kN (upward, the loads are symmetric)

Bending moment:

$0 \le x \le 0.2$: $M = 40x \rightarrow M(0.2) = 8$ kN·m

$0.2 \le x \le 0.8$: $M = 40x - 40(x - 0.2) = 8$ kN·m (constant; the shear force V = 0)

$0.8 \le x \le 1.0$: M decreases linearly back to 0 at x = 1.0



Critical region: 0.2 m ≤ x ≤ 0.8 m (largest M and T together)

(b) Maximum bending moment and internal torque

$M_{max} = R_A \cdot a = (40 \text{ kN})(0.2 \text{ m})$

M_{max} = 8 kN·m (constant between the two loads)

The two torques are equal and act in opposite directions. One is put into the shaft at x = 0.2 m and taken out at x = 0.8 m, so the shaft section **between the loads** carries the full torque. The end parts carry no torque.

$T_{internal}$ = 40 kN·m (for 0.2 m ≤ x ≤ 0.8 m)

(c) Minimum shaft diameter – von Mises

The critical section is in the middle region, where M = 8 kN·m and T = 40 kN·m act together and V = 0. The critical points are the top and bottom fibers.

$\sigma = 32M / \pi d^3 \quad \tau = 16T / \pi d^3$

$\sigma' = \sqrt{(\sigma^2 + 3\tau^2)} = (16 / \pi d^3)\cdot\sqrt{(4M^2 + 3T^2)} = S_y / n$

$\sqrt{(4M^2 + 3T^2)} = \sqrt{(4(8000)^2 + 3(40000)^2)} = \sqrt{(2.56\times10^8 + 4.80\times10^9)} = \sqrt{(5.056\times10^9)} = 71\ 106\ \text{N·m}$

$S_y / n = 400 / 2 = 200\ \text{MPa}$

$d^3 = 16\cdot n\cdot\sqrt{(4M^2 + 3T^2)} / (\pi\cdot S_y) = 16(2)(71\ 106) / (\pi \cdot 400\times10^6) = 1.8107\times10^{-3}\ \text{m}^3$

$d = (1.8107\times10^{-3})^{1/3} = 0.1219\ \text{m}$

d_{min} = 121.9 mm ≈ 122 mm

Check with d = 0.1219 m: σ = 45.0 MPa, τ = 112.5 MPa, so $\sigma' = \sqrt{(45.0^2 + 3\cdot112.5^2)} = 200\ \text{MPa} = S_y/n$ ✓. In practice choose the next standard size, e.g. **d = 125 mm**, which gives n = 2.16.